Plamen Markov



Senior Backend Engineer — Laravel / PHP · REST APIs · LLM Integration

plamen326@gmail.com · Pleven, Bulgaria · Europe/Sofia (UTC+2/+3)
github.com/plamen911 · linkedin.com/in/plamen-markov-14254965 · Fully remote · Available as a B2B contractor

PROFESSIONAL SUMMARY

Builds **LLM features into production software** and owns them after launch: a tool-calling agent that is never allowed to compute a figure itself, a Claude plugin held to an LLM-judge eval suite, an MCP server published on npm, and vision OCR. Before that, **two decades of remote backend work**, most of it as the engineer responsible for the whole system – one platform for 13 years, from architecture and data model to rollout and maintenance. PHP and Laravel are the foundation; TypeScript and LLM integration are where the work is heading.

SELECTED MEASURABLE RESULTS

18 years — my first commercial product (2008), deployed at 10+ client firms, still running

20 years — continuous remote engagement with one US software studio (2005–2025)

86.5% — of a 60,000-line delivery platform authored personally; its lead for 13 years

4 systems — modernised from PHP 5.6 to modern Laravel / PHP 8.2

26,000+ — insurance policies migrated live, cutover parity-tested

436 tests — introduced to a 3-year-old production SaaS that had almost none

PHPStan 9 — plus an 878-test suite, gating every push in CI

AI ENGINEERING

AI-assisted development — how I write code

Claude Code every day for over a year inside **client production codebases**, not side projects: AI-assisted refactoring, test generation and debugging. The scarce skill is not generating code — it is reviewing what was generated, at senior depth, before it reaches paying customers.

AI inside the product — what I build

Designed and shipped a **tool-calling LLM reporting agent forbidden from computing any figure itself**: every number comes from one of fifteen curated tools with enum-constrained schemas, so the model picks from a fixed filter catalogue instead of free-forming a query. Gated behind an authorisation check and a feature flag because it surfaces personal data, and kept off the public MCP endpoint. Also vision-based document OCR (17 fields from a vehicle registration) and two MCP servers, one on npm automating the motor-insurance register check (case study: github.com/plamen911/go-insurance-checker; npm: go-insurance-checker). **The value here is constraint design, not ML research.** Separately I built and deployed a **RAG** product of my own end to end — **fire-expert.uk**, below.

WORK EXPERIENCE

Smartwary, Bulgaria — Senior PHP Developer (B2B contract) — **Aug 2025 – present**

Remote. Built **Browswave**'s pricing-intelligence platform until the client took it in-house in October 2026 — market intelligence for the consumer-electronics industry. 489 commits in 13 months; roughly 45% of tracked source.

Ivanov New Media Studio LLC — Nantucket, MA, USA — PHP Web Developer (contractor) — **2005 – 2025**

Remote throughout. Twenty years as the studio's long-term remote developer, building and maintaining production software for its US clients — among them Nantucket Energy, whose fuel-delivery platform I led for thirteen years.

Direct clients — Independent contractor — **2008 – present**

Remote. Sole developer on three production systems, all still in service and 100% my code: **Hipokrat** (hipokrat.net), an occupational-medicine application written in 2008 and since deployed for 10+ client firms — web-hosted, on-premises and standalone; **NPP Kozloduy**'s occupational-health system (since 2012); **General Broker**'s policy platform (17+ years). I gather the requirements directly from the business people who use them.

KEY PROJECTS

Browswave — PriceLens & PromoManager · multi-tenant pricing SaaS — **2025 – 2026**

Laravel 10 · PHP 8.2 · MySQL 8 · stancl/tenancy · Shopify REST + GraphQL · Docker · AWS

- **Database-per-tenant across 24 tenant databases** — consumer-electronics retailers and Shopify merchants across Bulgaria and the UK; the largest holds 35,500 products, 83,700 competitor records and roughly 18 million rows of pricing history.
- **Shopify rate-limit resilience across the tenant fleet** — proactive leaky-bucket pacing from the GraphQL cost extensions and the REST call-limit header, *Retry-After*-honouring retries with jittered backoff capped at 32 s, idempotent webhooks deduped on webhook ID, batched bulk mutations.

General Broker — insurance platform · sole developer, 100% of the code

2009 – present

Laravel 13 · PHP 8.4 · Livewire 3 · MariaDB · Redis · laravel/ai · Symfony 7.3 · React 18/19 · Bitbucket Pipelines · GitHub Actions

- **Made one place own the money.** Browser and server rounding disagreed by a cent, enough to block saves, so every premium, tax and total now comes from one server endpoint. Saves are transactional, non-destructive; invariant checks repair only under an explicit flag, after a snapshot.
- **Shipped a circuit breaker for the outbound integrations** after an upstream served an expired certificate for ten weeks. Three states in one cache entry, a lock so exactly one probe runs; wired to that upstream first, behind a seam the others join one at a time. 14 tests.
- **Built the property-quoting service alongside it** — a separate *Symfony 7.3* API (20 Doctrine entities, 57 routes, JWT with anonymous tokens), a React 18 quote wizard and a React 19 admin console.
- **Quality gates on both codebases** — PHPStan level 9, 878 tests and dependency audits gate every push on the Laravel platform; the Symfony service's first suite, 219 tests, surfaced an authorisation gap reaching admin data.

My own products — fire-expert.uk, Claude plugin, ADR Zone · sole developer

2026

Laravel 12 · PHP 8.5 · PostgreSQL 17 + pgvector · OpenAI + Cohere · Claude skills · n8n · React Native · Expo

- **Claude plugin drafting court-grade fire reports — 5–6 h to 10–15 min** from photos of 15–20 key pages of a 45–70-page case file; LLM-judge evals, PII checks in code.
- **Built and deployed a RAG product end to end, alone** — a Bulgarian-language retrieval-augmented generation assistant that gathers case facts conversationally, retrieves comparable past reports and emits a structured court-format expert report. Deployed on a 4 GB VPS: Docker Compose, TLS, nightly *pg_dump* backups, scripted *git archive* deploys with smoke tests.
- **ADR Zone — React Native / Expo app, live on the App Store** (apps.apple.com/bg/app/id6769773386). Offline dangerous-goods reference over a bundled 1,818-substance SQLite database, searched by name, UN number or placard code. Relevance ranking added after a substring match buried pure chlorine behind hundreds of chloride compounds - in an emergency-response tool that is a safety property, not a UX nicety - and a Cyrillic case-fold layer, because Hermes' *toLowerCase* and SQLite's *COLLATE NOCASE* are both ASCII-only.

Nantucket Energy — propane & heating-oil delivery platform · principal developer

2012 – 2025

PHP 8.1+ · Laravel 9 · Livewire · MySQL 8 (112 tables) · Alpine.js · Docker · Larastan · AWS S3

- **Thirteen years, two systems.** I wrote the original platform from scratch, then built its Laravel replacement and the bridge between them. **~52,000 of ~60,000 surviving lines (86.5%) and 1,481 of 2,499 commits** — principal author, active every year the project ran, verified by line-level *git blame*.
- **Integrated four independent tank-telemetry vendors across four transport styles** — hand-built SOAP envelopes to a .NET endpoint, two session-authenticated REST APIs, and a LoRaWAN webhook decoding raw hex payloads into battery voltage and tank percentage — plus the scheduled batch ingestion behind them.

NPP Kozloduy — occupational-health & medical-surveillance system · sole developer

2012 – present

PHP 8.2 · MySQL/MariaDB · Symfony components · Monolog · Pest · PHPStan · air-gapped intranet

- **Rebuilt in place from PHP 5.6 to PHP 8.2 with no rewrite and no flag-day cutover** — 114 domain services extracted from 172 flat page scripts, DTOs with Symfony Validator, Monolog and Pest. A parallel Laravel rewrite was started, evaluated and **rejected**, with the reasoning recorded in four ADRs including an explicit “no framework migration” decision.

TECHNICAL SKILLS

Backend	PHP 8.x (20+ yrs) · Laravel (expert) · **Symfony 7** (production API) · Doctrine ORM · Livewire · Node.js
Data	MySQL / MariaDB (expert) · **PostgreSQL 17 + pgvector** (vector similarity search) · query-plan & index diagnosis · Redis · SQLite · MongoDB (basic)
AI	Claude Code in production (daily, 1+ yr) · **RAG** (pgvector + full-text hybrid, Cohere rerank) · laravel/ai · MCP servers · tool-calling agent design · OpenAI vision OCR
Integrations	REST · GraphQL · SOAP · JSON-RPC 2.0 · webhooks · SFTP · SAML/OAuth flows · rate-limit resilience · Shopify · Amazon (Keepa)
Resilience	**circuit breaker (shipped)** · leaky-bucket rate-limit pacing · jittered backoff · idempotent webhooks · deadlock diagnosis & retry · queues · heartbeat alerting
Testing & quality	PHPUnit · Pest · Laravel Dusk · PHPStan / Larastan to level 9 · Pint · CI pipelines with dependency-audit gates
Mobile	**React Native 0.81** · Expo SDK 54 · TypeScript · offline SQLite · EAS build & submit (App Store)
Frontend & infra	Livewire · Alpine.js · **React 18/19** (two production front ends) · **Vue 2** · **Svelte 4** · Tailwind · Docker · AWS (EC2, S3) · nginx / php-fpm · Git

EDUCATION

SoftUni — PHP Web Developer, 2019 (894 h); Front-End Developer with JavaScript, 2019 (289 h)

TU Gabrovo — Computer Systems & Technologies, 2005

LANGUAGES & AVAILABILITY

Bulgarian — native · English — professional working proficiency

B2B via a VAT-registered BG company; up to full-time.